

EXP NO: 3

Date: 21/03/23

SQL COMMANDS FOR NESTED QUERIES

AND JOIN QUERIES

Aim:

To execute and verify the SQL commands using nested queries and join queries.

Procedure:

Step 1: Start

Step 2: Create table with its essential attributes

Step 3: Insert attribute values into table.

Step 4: Execute different commands and extract information from the table.

Step 5: Stop.

SQL COMMANDS

1. COMMAND NAME: INNER JOIN

COMMAND DESCRIPTION: INNER JOIN Command returns the matching rows from the tables that are joined.

Select: $e.$ Employee-Id, $e.$ first-name, $e.$ dept-^{Id}name, $d.$ dept-name
from Emp1 $\text{Inner join Dept1 on } e.\text{dept-Id} = d.\text{dept-Id};$

2. COMMAND NAME: LEFT OUTER JOIN

COMMAND DESCRIPTION: LEFT OUTER JOIN command returns matching rows from the tables being joined and also non-matching row from left table in result and places null values in the attribute that come from right table.

Emp1:

Emp-id	First_name	Last_name	Job-id	Salary	Man-id
101	Ramya	Couppin	1010	200000	2424
102	Karthik	Kumar	0606	20000	1212
103	Abhinaya	Sudhagar	1111	50000	6060
104	Tharu	thiru	0101	60000	1717

Dept:

Dept_id	Dept_name	Manager_id	Location_id
1	CSE	1212	605012
1	CSE	2424	605015
3	ECE	1717	605015

INNER JOIN:

EMP-ID	FIRST NAME	DEPT-ID	DEPT. NAME
101	Ramya	1	CSE
102	Karthik	1	CSE
103	Thanu	3	ECE

LEFT OUTER JOIN:

Emp-ID	First name	Dept-id	Dept-name
101	Ramya	1	CSE
102	Karthik	1	CSE
103	Thanu	3	ECE
104	Abhiramya	2	

RIGHT OUTER JOIN:

Emp-Id	First-name	Dept-id	Dept-name
101	Ramya	1	CSE
102	Karthik	1	CSE
104	Thanu	3	ECE

FULL OUTER JOIN:

Emp-Id	First-name	Dept-id	Dept-name
101	Ramya	1	CSE
102	Karthik	1	CSE
103	Abhiramya	2	
104	Thanu	3	ECE

Query: $\text{Select } e.\text{employee_id}, e.\text{first_name}, e.\text{dept_id}, d.\text{dept_name}$ from
 $\text{emp } e \text{ left Outer join dept } d \text{ on } e.\text{dept_id} = d.\text{dept_id};$

3. COMMAND NAME: RIGHT OUTER JOIN

COMMAND DESCRIPTION: RIGHT OUTER JOIN command returns matching rows from table being joined and also non matching row from the right table in result and places null values in the attribute that come from left side table

Query: $\text{Select } e.\text{employee_id}, e.\text{first_name}, e.\text{dept_id}, d.\text{dept_name}$ from
 $\text{emp } e \text{ Right join dept } d \text{ on } e.\text{dept_id} = d.\text{dept_id};$

4. COMMAND NAME: NESTED QUERY

COMMAND DESCRIPTION: NESTED QUERY command runs query within another query.

Query: $\text{Select } e.\text{employee_id}, e.\text{first_name}$

5. COMMAND NAME: FULL OUTER JOIN

Query: $\text{Select } e.\text{employee_id}, e.\text{first_name}, e.\text{dept_id},$
 $d.\text{dept_name}$ from $\text{emp } e \text{ full join dept } d \text{ on}$
 $e.\text{dept_id} = d.\text{dept_id};$

Result:

Thus the SQL Command for Nested Queries and Join Queries was executed successfully.